Reducing Choice Overload in Amazon E-commerce platform

Submitted by: Sourish Chatterjee

What is decision fatigue?

Decision fatigue in the context of products refers to the mental exhaustion and difficulty consumers experience when faced with too many choices, leading to indecisiveness and potentially poor decisions.

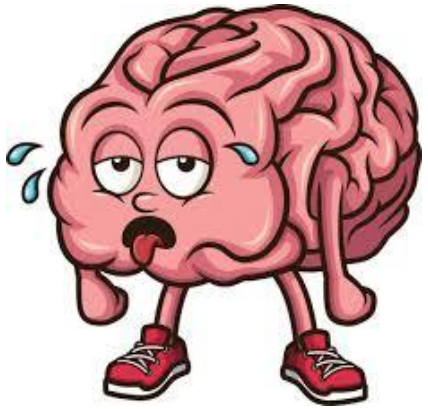
The products we use often run the risk of offering too many choices for its customers. As a result, users feel:

- **Overwhelmed by Options:**

Consumers can become overwhelmed by the sheer number of products, features, or options available, leading to a feeling of being paralyzed by choice.

- **Mental Exhaustion:**

Making numerous decisions, especially when they are complex, can drain mental energy, leading to decision fatigue.



Impact on Purchasing Decisions

Decision fatigue can lead to:

- **Indecisiveness:** Consumers may become hesitant or unable to make a choice.
- **Impulse Purchases:** In a state of fatigue, consumers may make quick, potentially regretted purchases.
- **Delayed Purchases:** The mental effort required to evaluate options can lead to postponing the purchase decision.
- **Suboptimal Decisions:** When mental resources are depleted, consumers may make choices that are not ideal or that don't fully meet their needs.

This ultimately leads to a poor experience with the app and leads to increased poor retention and user drop-off rates.

Impact on User Experience

When a product's interface or features are too complex or require too much mental effort to understand, users experience cognitive overload. This can lead to:

- **Frustration:** Users struggle to complete tasks, leading to feelings of helplessness and annoyance.
- **Abandonment:** Users may give up on using the product altogether if it's too difficult to navigate or use.
- **Poor Retention:** Users are less likely to return to a product that they found difficult or confusing to use.

Is it a real Problem?

- **35%** Online shoppers abandon their carts due to difficulty finding what they need
- **44%** Users feel filters lack a helpful degree of specificity
- **75%** Consumers are do not trust user reviews on the platform
- **35%** Shoppers struggle with search filters, complicating product discovery.
- **56%** Visitors drop off before viewing a product details page, overwhelmed by choices

Primary reason: Too many choices for user

Secondary reasons:

- **Inefficient filter options**
- **Untrustworthy paid reviews**

Traditional Ways of buying

Customer goes to the local market → Goes to a suitable shop → Interacts with the shopkeeper regarding his need → Buys vendor's suggestion → No decision fatigue

Pros

- Almost all the cognitive load of choosing a good product is outsourced to the vendor
- Local vendors are trusted by customers

Cons

- Local vendor/marketplace can be far and difficult to commute to
- Time consuming be available
- The desired items may not be always available
- Home delivery may or may not be an option

Amazon flow for single product requirement

Opens app → Searches for requirement → Applies filter → Selects suitable product → Reads reviews

Dislikes product → Searches for requirement

Likes product → Selects variations, if any → Adds to cart → Buys → High decision fatigue

Pros

- Gets a wide range of products available
- Eliminates need to travel
- Discounts options available

Cons

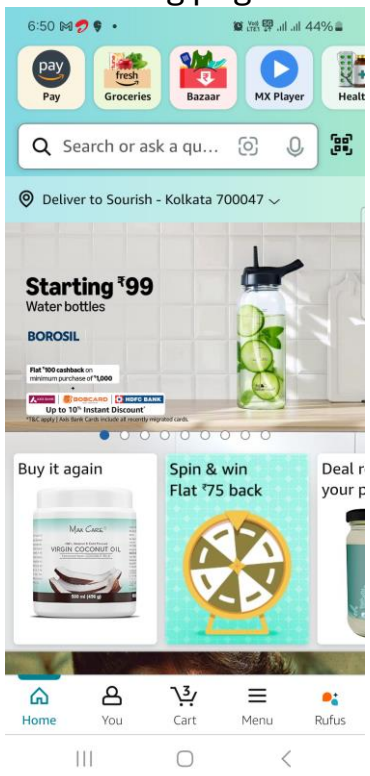
- Too many variations of a single product
- Reviews difficult to trust
- Selecting takes time
- Entire process manual, customers do all the heavylifting of applying correct filters, reading reviews and choosing the product takes time
- Entire process become very tedious for multiple products

User Segmentation

	User Type	User Attributes	User Story	Challenges faced	Requirements
	Value-Seeking Shopper	Name: Rajesh, Age: 25-40, Profession: Office Employee, Status: Married, Income: ₹3-6L, Type: Budget-Conscious	Orders during sales, spends time comparing deals & discounts, wants more in a short time before products run out of stock	Too many deals & variations make it hard to identify real savings.	Requires a product with the best deals in a short time .
	Premium Shopper	Name: Priya, Age: 22-45, Profession: IT Professional, Status: Single/Married, Income: ₹8-20L, Type: Prime Subscriber	Limited time but orders high-value items, prefers fast ordering, expects same/next-day delivery.	Gets frustrated when when options are unclear and decision takes time or longer delivery times	Personalized recommendations, to buy fast and buy more
	Tech-Savvy Buyer	Name: Ankit, Age: 18-35, Profession: Engineer/Student, Status: Single, Income: ₹5-12L, Type: Gadget Enthusiast	Orders as soon as new products launch, reads/watches multiple reviews before buying	Overloaded with reviews & specs, making decision-making difficult	Notifications when new tech is available for sale, or when price drops
	Fashion Shopper	Name: Sakshi, Age: 18-35, Profession: Student/Professional, Status: Single, Income: ₹3-10L, Type: Trend Seeker	Buys during festive sales, browses long before purchasing	Too many choices create indecision & product can run out-of-stock before buying	Automatic selection of her preferred sizes/materials and suggestions based on buying history
	Business Buyer (B2B)	Name: Arvind, Age: 30-55, Profession: Business Owner, Status: Married, Income: ₹10-25L, Type: Bulk Buyer	Buys in bulk, needs predictable restocking options, wants the best value in terms of price and quality	Pricing fluctuations & bulk deals are hard to track.	Locking price, and to see only products with bulk options and subscribing available

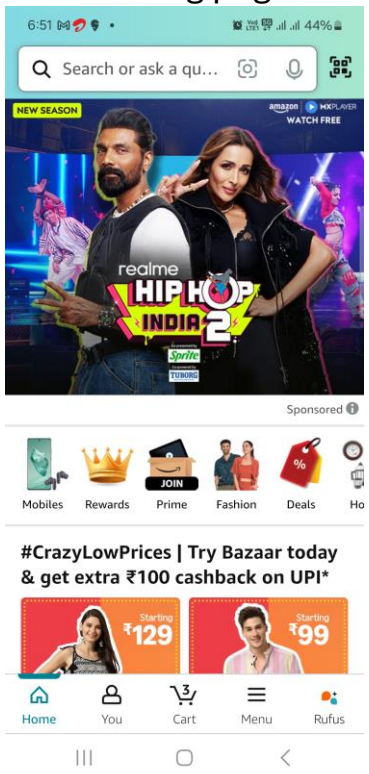
Current Scenario

Landing page



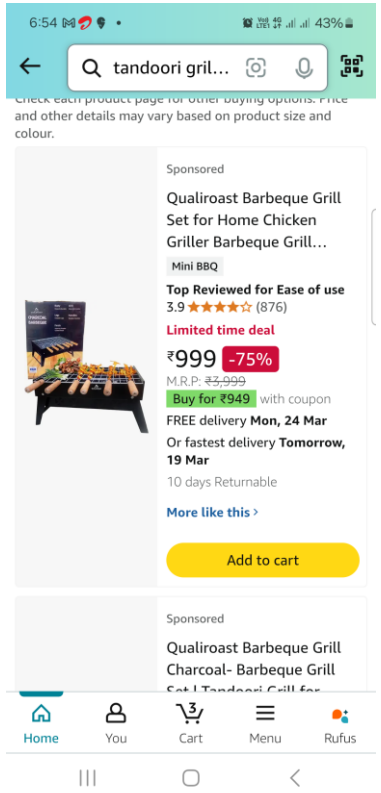
- Categories missing
- Very little relevant substance
- 1 offer taking up half the page

Landing page



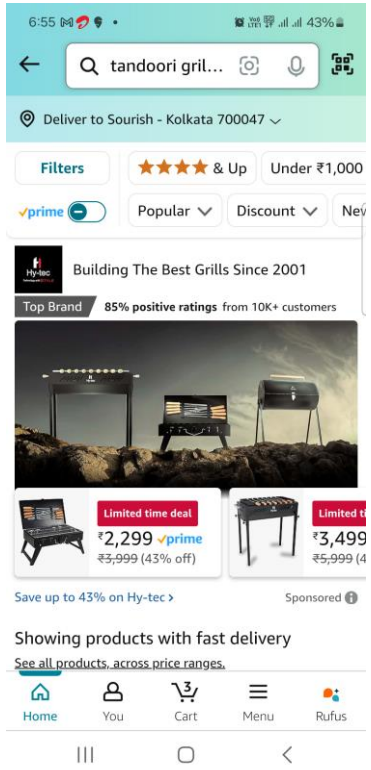
- Irrelevant ads
- Personalized recommendations missing

Search

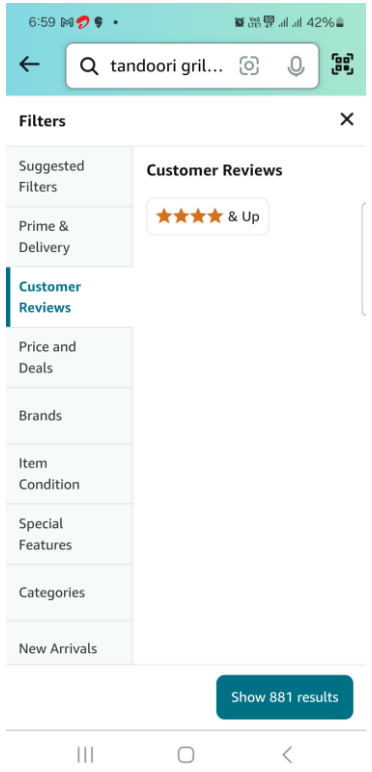


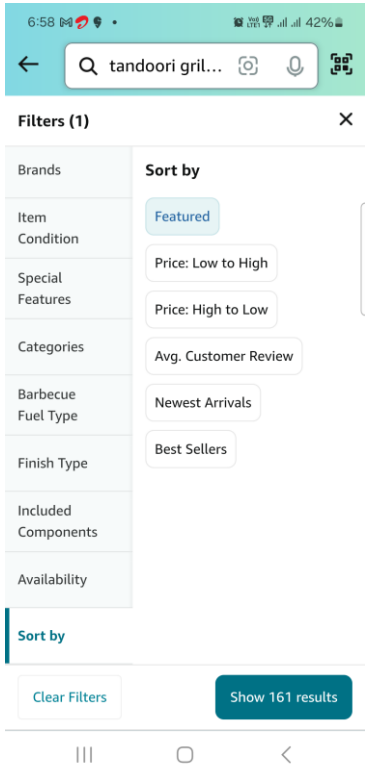
- One frame has just one item = More scrolling
- Top reviewed has lower rating = Less trust

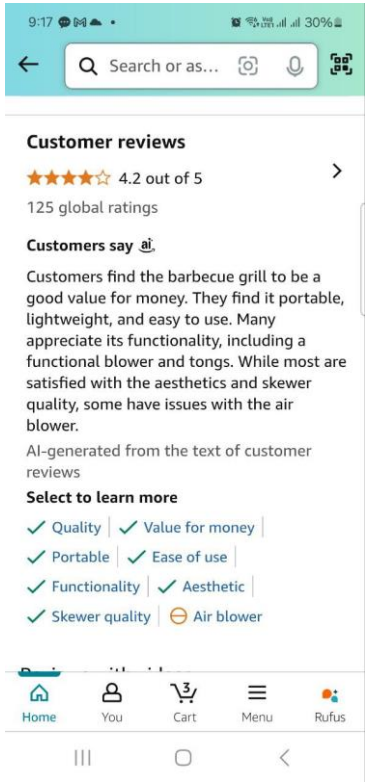
Select



- Distracting graphics
- Number of results not displayed







- Useful filter options like number of reviews missing.
  4.2 rating(1k reviews > 5 rating(5 reviews)

- Limited sorting options

- Lots of scrolling before reviews are visible
- Very very exhaustive non-collapsible product decription and other similar products in between

Breakdown

- **Time**: Too much time to narrow down to a single relevant product
- **Distraction**: Irrelevant ads utilizing maximum space
- **Sponsored products**: showing up on top of better options
- **Filters**: Completely Manual
- **Space**: Inefficient utilization with exhaustive desciptions

Business Impact

Increased Cart Abandonment Rate

- Shoppers overwhelmed by too many choices leave without completing purchases
- Revenue loss & lower conversion rates

Lower Customer Satisfaction & Retention

- Unclear filters and fake reviews create negative brand perception and lower retention

Reduced Engagement & Repeat Purchases

- Decision-making gets tiring and shopping frequency reduced
- Lower customer lifetime value (CLV) for Amazon

Increased Return Rates

- Impulse decisions leads to higher returns and higher return costs for Amazon

Greater Dependency on Promotions & Discounts

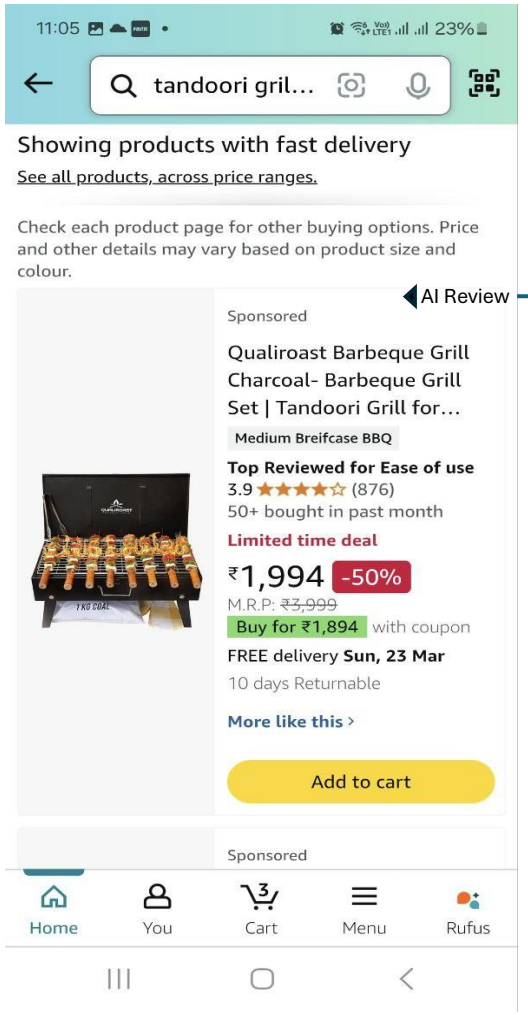
- Users rely on sales like Amazon Great Indian Festival to reduce choice complexity and skews profits and shrinks organic purchase

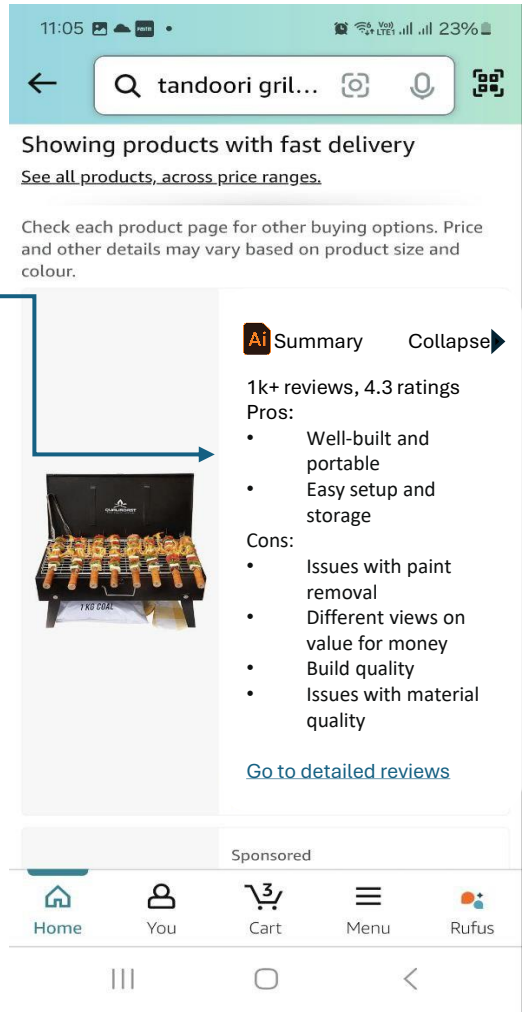
Solutions and Prioritization

Feature	Functional desciption	Non-Functional desciption	Reach	Impact	Effort	Net Score
Quick access for review tab	A right arrow on the top left corner of the product listing, collapsible in horizontal direction	AI summary of reviews to include number of people reviewed, pros and cons	L	L	S	9
AI prompt search	A separate category in the filters tab	Selects the relevant filters automatically mentioned in the tab	L	L	M	4.5
Categorization	Separate section on the landing page	Proper categorization helps users to directly move to what they are looking for	M	M	S	4
Paid Ads free model	Subscription based model with Prime	Allows us to retain high-value customers while enhancing their experience	M	L	M	3
Personalized recommendation	"Recommended for You" on every search, horizontally scrollable, a list of 5 most relevant items based on purchase history	Allows users to directly choose a proper option	M	L	L	2
Better Filter options: Filter by Image, Number of reviews	Separate sections on the top of filter options	Better and more comprehensive filters for efficient searching	M	M	M	2

Solution Analysis

Quick access for review tab





- **Functional role**: Collapsible ribbon that shows AI summary of reviews for the product
- **Non-functional role** : Should be able to analyse 2k reviews for 1k products on a single search

- Ai summary gives a concise summary of number of reviews and pros and cons of all the reviews
- **'Go to detailed reviews'** quickly skips to the reviews page

User Impact

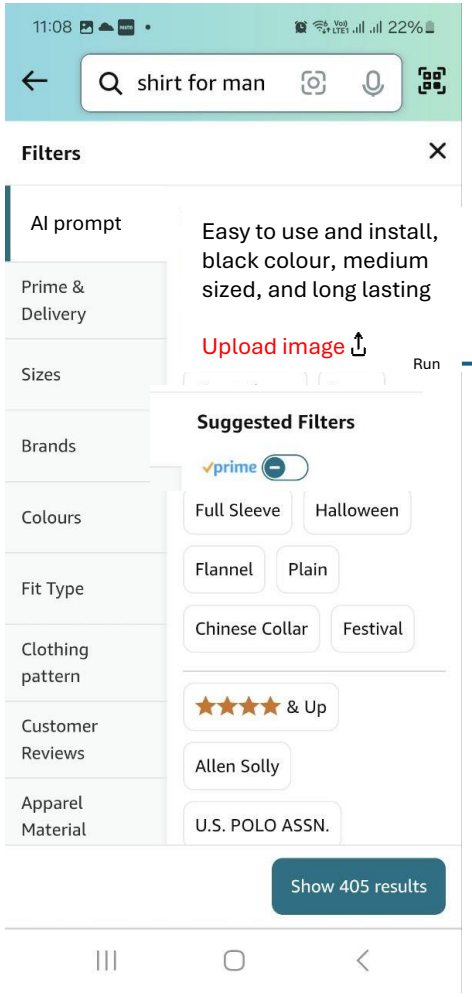
- **Faster Decision-Making** – Quickly available reviews skips acrolling to multitudes of data and provides a wuick snapshot of what others have to say about the picture.
- **Minimal scrolling required**– Saves a lot of time in scrolling and analysing the product

Business Impact

- **Higher Conversion Rates** – Users make decisions faster, leading to increased purchases
- **Lower Cart Abandonment** – Eliminates the frustration of analysis paralysis, reducing drop-offs
- **More Engagement** – A one-click AI summary keeps users within the Amazon ecosystem, reducing time delay in scrolling through tabs
- **Trust & Retention Boost** – AI-backed summaries of verified purchases increase confidence in purchases, improving customer satisfaction and repeat buying behavior.

Solution Analysis

AI Prompt filter



'Run' button executes prompt

- **Functional role:** Separate option in Filter tab, with text box to insert AI prompt. Should also include 'Upload Image' options to enable better prompt search
- **Non-functional role** : Should be able to analyse 2k products and reviews to match the prompt. Max 2 images can be supported

User Impact

- **Personalized product discovery** – AI can tailor analysis based on user preferences (e.g., budget shoppers see price-value insights first)
- **Faster & Smarter Product Discovery:** Users no longer need to manually filter through thousands of listings.
- **Contextual Recommendation**: AI understands intent beyond keywords (e.g., "best laptop for college students" filters for lightweight, good battery, mid-range price or "show only 4 stars and above").
- **Personalized Recommendations:** AI factors in user preferences, past purchases, and browsing history for hyper-relevant results
- **Accessibility & Ease of Use:** Helps non-tech-savvy users who struggle with complex filters, supports image search prompts for reference product discovery

Business Impact

- **Higher Conversion rate –** Faster, intent-based product discovery reduces drop-offs and increases purchase completion
- **Better Ad Targeting** – AI-driven searches also hints better at what exactly a customer wants, this improves relevance for sponsored listings, enhancing ad performance.
- **Optimized Seller Visibility –** Surfaces niche and new sellers based on user intent rather than ad spend dominance.
- **Lower Cart Abandonment –** Eliminates the need for external research and manual filtering
- **Higher Retention & Engagement** – Keeps users within Amazon's ecosystem by making product discovery seamless and personalized

Metrics

NSM

- **Increase in Conversion Rate–** Measures the change in percentage of users who complete a purchase after searching for a product

Primary Metrics

- **Click-Through Rate (CTR) on AI Results –** Measures engagement with AI-suggested products
- **Seller Impression Share –** Measures visibility of new and niche sellers in search results
- **Average Order Value (AOV) –** Tracks if AI suggestions lead to higher cart value

Primary Metrics

- **Drop-off rate from AI Search–** Tracks if users leave Amazon after using AI search, indicating irrelevant results**Average**
- **Reduction in Organic Review Contributions –** Measures if users rely too much on AI summaries, leading to fewer reviews

thank you

FOR YOUR ORDER